

QAEMSHAHR, Iran, Islamic Republic of

WMO: 407370

Lat: 36.450N

Lon: 52.817E

Elev: 46

StdP: 14.67

Time Zone: 3.50 (IRN)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.8	33.2	27.8	21.8	34.5	30.4	24.6	36.7	19.2	46.2	17.4	48.1	2.1	180	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.9	93.7	78.5	91.6	77.5	89.7	76.9	81.0	89.9	79.8	88.2	78.4	86.5	7.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.7	149.0	86.7	77.3	142.4	85.3	76.1	136.4	83.7	44.9	89.9	43.4	88.2	42.1	86.9	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	13.5	11.6	DB	27.3	98.8	1.7	2.4	26.1	100.5	25.1	101.9	24.1	103.3	22.9	105.0	
			WB	26.3	83.1	2.5	1.7	24.5	84.3	23.1	85.3	21.7	86.2	19.9	87.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	45.8	46.8	52.6	59.9	70.0	76.8	79.8	81.4	75.8	66.5	55.8	48.9
	DBStd	13.68	5.03	4.83	5.92	6.04	5.05	3.56	3.65	3.48	4.18	5.16	6.05	4.80
	HDD50	386	146	108	32	3	0	0	0	0	0	0	17	80
	HDD65	2503	594	508	389	175	13	0	0	0	0	43	281	500
	CDD50	5291	17	20	113	298	620	805	925	973	773	511	191	45
	CDD65	1934	0	0	5	21	168	355	460	508	323	89	5	0
	CDH74	16824	0	2	57	223	1044	2889	4360	5380	2377	458	32	2
	CDH80	6234	0	0	17	76	285	946	1678	2416	724	87	5	0
Wind	WSAvg	4.9	4.6	4.9	5.6	5.3	5.4	5.2	4.9	4.9	4.8	4.2	4.1	4.4
Precipitation	PrecAvg	33.2	2.6	2.6	3.1	2.0	1.5	1.7	2.0	1.9	4.3	4.1	4.7	2.8
	PrecMax	63.3	5.7	5.4	6.5	4.4	2.9	5.4	5.0	5.2	14.0	9.9	8.3	4.8
	PrecMin	25.1	0.5	0.7	1.0	0.0	0.5	0.5	0.5	0.1	0.8	1.6	0.9	0.8
	PrecStd	7.6	1.2	1.2	1.2	1.1	0.6	1.1	1.2	1.2	2.5	2.1	1.7	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.3	70.1	81.3	89.8	91.3	91.8	96.4	96.4	93.0	87.3	78.3	71.1
		MCWB	55.1	55.9	65.5	69.9	74.2	76.3	79.3	78.3	76.7	71.1	64.7	58.9
	2%	DB	62.1	63.4	73.3	79.4	86.2	89.4	92.8	94.1	89.4	81.9	72.0	64.6
		MCWB	53.0	53.2	59.6	66.0	72.2	75.8	79.0	78.8	76.5	70.2	62.7	55.4
	5%	DB	57.6	58.4	67.9	74.6	83.1	87.4	90.3	92.3	86.6	78.9	68.5	60.5
		MCWB	50.6	51.4	58.4	64.1	70.9	75.1	77.7	78.4	75.6	69.0	61.0	54.1
	10%	DB	54.3	54.9	62.6	70.3	80.0	85.4	88.1	90.5	84.2	76.0	65.8	57.5
		MCWB	49.1	49.7	55.6	62.4	69.5	74.3	76.7	77.6	74.2	67.9	60.1	53.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.0	57.8	66.4	70.7	76.3	80.5	83.4	82.7	80.7	74.0	67.4	59.9
		MCDB	66.0	67.6	79.7	83.5	87.2	88.5	92.5	92.1	89.0	82.2	74.3	67.4
	2%	WB	53.7	54.4	61.9	67.4	74.2	78.2	80.9	81.3	78.5	72.0	64.5	57.2
		MCDB	60.3	61.2	71.0	78.0	83.7	86.3	89.6	90.4	86.2	79.1	69.6	62.1
	5%	WB	51.4	52.1	58.5	64.6	72.0	76.6	79.2	80.1	76.9	70.4	62.7	55.3
		MCDB	55.9	56.8	66.4	73.5	80.4	84.4	87.1	88.9	83.7	77.0	66.9	59.7
	10%	WB	49.8	50.4	55.8	62.6	70.4	75.2	77.9	78.9	75.6	68.9	60.6	53.5
		MCDB	53.4	53.9	62.3	69.1	78.2	83.0	85.3	87.5	82.0	74.6	64.8	56.8
Mean Daily Temperature Range	5% DB	MDBR	16.8	14.8	14.9	15.4	15.0	14.4	13.5	15.9	14.4	15.8	16.0	16.2
		MCDBR	23.3	24.0	26.1	25.1	21.4	18.0	17.9	19.9	18.0	20.5	21.2	22.2
	5% WB	MCWBR	15.3	15.2	15.3	13.6	10.4	8.1	7.8	8.8	8.2	10.4	12.6	14.9
		MCDWR	20.1	20.3	24.4	22.0	19.1	16.3	16.4	17.5	16.1	17.2	17.0	19.1
Clear-Sky Solar Irradiance	taub		0.385	0.429	0.489	0.562	0.573	0.577	0.623	0.583	0.552	0.511	0.438	0.390
	taud		2.212	2.089	1.956	1.833	1.854	1.876	1.799	1.879	1.927	1.986	2.117	2.206
	Ebn at Noon		248	250	246	235	234	233	221	227	225	220	225	236
	Edh at Noon		35	44	55	65	65	63	68	62	56	48	38	33
All-Sky Solar Radiation	RadAvg		747	928	1213	1513	1831	2031	1936	1838	1482	1136	809	661
	RadStd		63	105	107	173	106	140	161	207	93	80	71	67

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	+0.45	N/A	-1.48	+1.10	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page